

AE5033 Flight Vehicle Simulator System

MODULE 6

Final Assignment

NAME : _____

SID : _____

Undergraduate Program in Aerospace Engineering

Faculty of Mechanical and Aerospace Engineering

INSTITUT TEKNOLOGI BANDUNG

Worksheet

Take-off maneuver is defined as the process of an aircraft to progress from a standing still on a runway to airborne at a certain height above it. The process involves changing engine settings and aircraft configurations, as well as a specific pilot input to achieve the desired final condition. The take-off process is depicted in Figure 1. The actual take-off process is considered to be completed when the best climb speed is achieved at the specified height (Climb (3) mark).

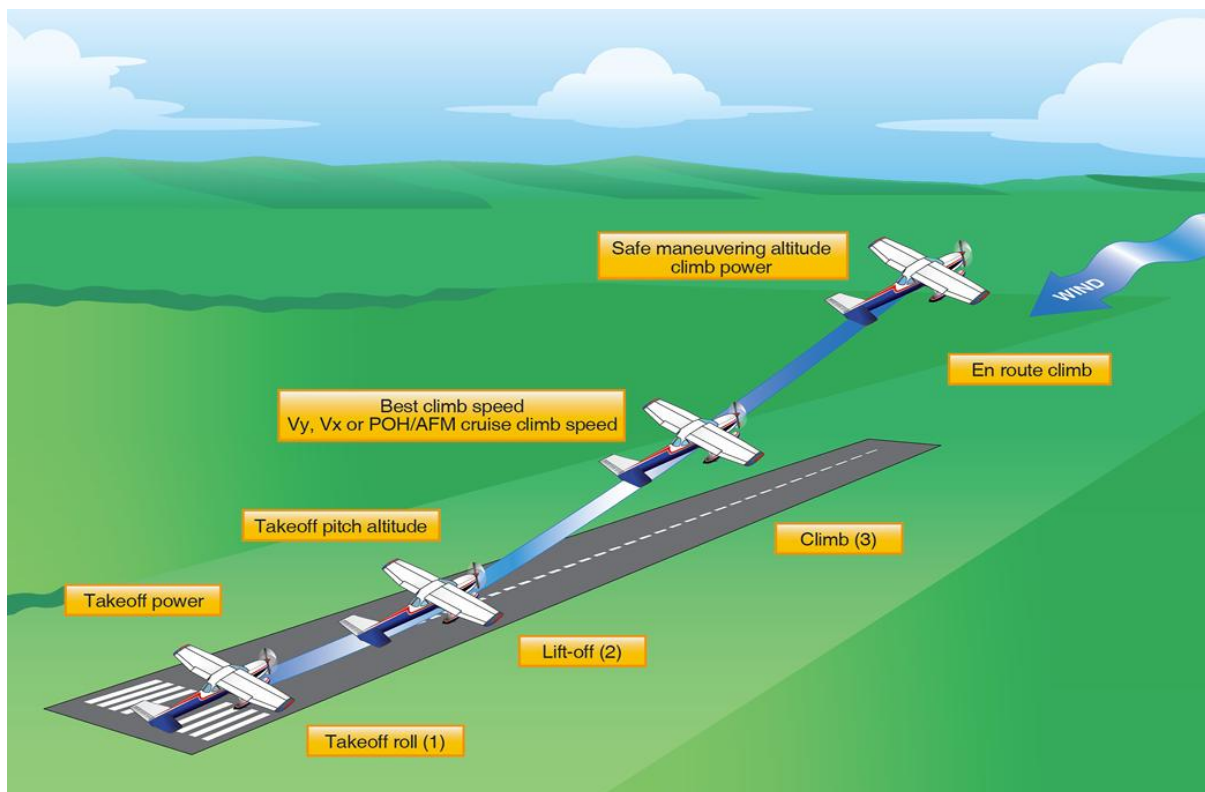


Figure 1. Illustration of Take-off maneuver up to the safe maneuvering altitude [1].

Take-off distance is defined as the distance required to complete the take-off maneuver from the standing still position. To simplify the analysis, the take-off maneuver is broken down into three stages:

1. Take-off ground roll
2. Lift-off
3. Initial climb

The take-off ground roll is the process of accelerating the aircraft from standing still into a speed safely beyond the stall speed where the aircraft begins the rotation sequence to gain sufficient lift to overcome its weight. Once the lift is equal to the weight of the

aircraft, the aircraft will lift off the ground and change the moving path from parallel to the runway to a desired climb angle.

In this module, you will be asked to perform take-off maneuvers for your selected aircraft and calculate the take-off distance for the aircraft. Please read the instructions and ask the assistant if there is ambiguity in the instructions.

Note: For this module you will need the data from the simulation operation. Please ask the assistant to ensure that the data recording is activated prior to your simulation.

TAKE-OFF DISTANCE DETERMINATION

1. Choose one of the aircraft from the available database.
2. Find manual document for the aircraft especially on the take-off reference speeds.
3. Ask the assistant to place the aircraft directly in the runway threshold, ready for take-off.
4. Set the flap configuration for take-off.
5. Proceed to take-off maneuver using your preferred technique (e.g. apply brake while increasing throttle to maximum).
6. If the speed has reached the rotation speed, pull up the stick gently to achieve the desired lift.
7. Hold on the stick once the pitch is about 10 degrees nose up.
8. Repeat the process of point 3 to 7, at least 10 times.
9. Take note of what you specifically do for each test run in the space below. (Note the rotation speed, flap settings, the resulting pitch angle after rotation, etc.)

Note :

Report Writing

For the report, please include the following:

1. Provide fundamental theory of take-off distance calculation.
2. Plot of elevator deflection vs time (All 10 maneuvers).
3. Plot the altitude response vs time. (All 10 maneuvers)
4. Plot the airspeed response vs time. (All 10 maneuvers)
5. Plot the distance traveled vs time. (All 10 maneuvers)
6. Using the definition, calculate the take-off distance for all 10 maneuvers.
7. What are the average, maximum, and minimum take-off distance.
8. Analyze what the cause of the differences are.
9. Provide suggestions on which take-off distance should be use in the Aircraft Flight Manual, and specify the procedure on how a pilot will be able to achieve the stated take-off distance.

Reports should be submitted at the deadline given on the Edunex. Delayed submission is allowed with some penalties on the the final mark.